# Jacob (Jake) Carlson

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https://jscarlson.github.io

**EDUCATION**

**Harvard University**, Cambridge, Massachusetts
Ph.D. Economics, August 2022 – May 2027 (expected)

**Yale University**, New Haven, Connecticut
B.A. Statistics and Data Science, August 2016 – May 2020
B.A. Ethics, Politics, and Economics, August 2016 – May 2020
*Summa cum laude* with distinction in both subjects

**HONORS AND AWARDS**

- Rising Stars in Data Science, University of Chicago, 2026
- PhD Fellowship Finalist, Two Sigma, 2026
- Presidential Scholar, Harvard University, 2022
- Undergraduate Thesis Prize Winner, Forethought Foundation for Global Priorities Research, 2020
- Phi Beta Kappa, Yale University, 2020

**WORKING PAPERS**

- Carlson, Jacob. "Interpretable Discovery from Unstructured Data: A High-Dimensional Approach." Working paper (2026).
- Carlson, Jacob and Neil Shephard. "When are time series predictions causal? The potential system and dynamic causal effects." Working paper (2026).
- Carlson, Jacob and Melissa Dell. "A Unifying Framework for Robust and Efficient Inference with Unstructured Data." Working paper (2025).

**CONFERENCE PUBLICATIONS**

- Carlson, Jacob, Tom Bryan, and Melissa Dell. "Efficient OCR for Building a Diverse Digital History." *ACL* (2024). **Oral presentation**.
- Dell, Melissa, Jacob Carlson, Tom Bryan, Emily Silcock, Abhishek Arora, Zejiang Shen, Luca D'Amico-Wong, Quan Le, Pablo Querubin, Leander Heldring. "American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers." *NeurIPS Datasets and Benchmarks Track* (2023).
- Bryan, Tom, Jacob Carlson, Abhishek Arora, Melissa Dell. "EfficientOCR: An Extensible, Open-Source Package for Efficiently Digitizing World Knowledge." *EMNLP System Demonstrations* (2023).
- Shen, Zejiang, Ruochen Zhang, Melissa Dell, Benjamin Lee, Jacob Carlson, and Weining Li. "LayoutParser: A Unified Toolkit for Deep Learning Based Document Image Analysis." *ICDAR* (2021). **Oral presentation**.

**JOURNAL PUBLICATIONS**

- Andrews, Isaiah, Harvey Barnhard, and Jacob Carlson. "True and Pseudo-True Parameters." Forthcoming, *Econometric Theory*.

- Carlson, Jacob, Trevor Incerti, and P.M. Aronow. "Dyadic Clustering in International Relations." *Political Analysis* (2024).

**EMPLOYMENT**

**Harvard University**, Cambridge, Massachusetts
Data Science Fellow in Deep Learning September 2020 – May 2022

- Developed deep learning methods and tools for social science data curation with Melissa Dell

**TEACHING**

**Harvard University**, Cambridge, Massachusetts

- *Causal Inference in Economics* (ECON 2148, Graduate): Spring 2025, 2026
- *Time Series* (STAT 242, Graduate): Fall 2024, 2025
- *Unleashing Novel Data at Scale* (ECON 2355, Graduate): Spring 2021

**INVITED TALKS**

- **2026**: OpenAI AGI Economics Conference (OpenAI, September); ESIF Economics and AI+ML Meeting (Cornell, June); AI and Economics Summer Conference (UChicago Booth, August)
- **2022**: ASSA Annual Meeting (Virtual, January)

**SOFTWARE**

- hdmht-discovery: `https://github.com/jscarlson/hdmht-discovery`
- mar-s (with Melissa Dell): `https://github.com/jscarlson/mar-s`
- NetworkTargeting (with Davide Viviano): `https://github.com/dviviano/NetworkTargeting`
- EffOCR (with Tom Bryan and Melissa Dell): `https://pypi.org/project/efficient-ocr`
- dcr (with Trevor Incerti and P.M. Aronow): `https://github.com/jscarlson/dcr`, `https://github.com/jscarlson/stata-dcr`

**ACADEMIC SERVICE**

*Referee*: Journal of Political Economy

**LAST UPDATED**

October 30, 2026